7. Prove that $\begin{vmatrix} \frac{1}{a} & bc & b+c \\ \frac{1}{b} & ca & c+a \\ \frac{1}{c} & ab & a+b \end{vmatrix} = 0$

8. Prove that $\begin{vmatrix} -a^2 & ab & ac \\ ab & -b^2 & bc \\ ac & bc & -c^2 \end{vmatrix} = 4a^2b^2c^2$

1.2 Inverse of a Matrix

1.2.1 Singular matrix:

A Square matrix A is said to be singular, if $|A| = 0$

1.2.2 Non – singular matrix:

A square matrix A is said to be non – singular, if $|A| \neq 0$

Example 1.10

Show that $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ is a singular matrix.

Solution

$$\text{Let } A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix}$$

$$= 4 - 4 = 0$$

$\therefore$ A is a singular matrix.

Example 1.11

Show that $\begin{bmatrix} 8 & 2 \\ 4 & 3 \end{bmatrix}$ is non – singular.

Solution

$$\text{Let } A = \begin{bmatrix} 8 & 2 \\ 4 & 3 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 8 & 2 \\ 4 & 3 \end{vmatrix}$$

$$= 24 - 8 = 16 \neq 0$$

$\therefore$ A is a non-singular matrix

DO YOU KNOW?

If A and B are non – singular matrices of the same order then AB and BA are also non – singular matrices of the same order.

1.2.3 Adjoint of a matrix

The adjoint of a square matrix A is defined as the transpose of a cofactor matrix. Adjoint of the matrix A is denoted by *adj A*

i.e., adj $A = [A_{ij}]^T$ where $[A_{ij}]$ is the cofactor matrix of A.

Example 1.12

Find adj A for $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$

Solution

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$$

$$\text{adj } A = [A_{ij}]^T$$

$$= \begin{bmatrix} 4 & -3 \\ -1 & 2 \end{bmatrix}$$

NOTE

(i) $|adjA| = |A|^{n-1}$ n is the order of the matrix A

(ii) $|kA| = k^n|A|$ n is the order of the matrix A

(iii) $adj(kA) = k^{n-1}adjA$ n is the order of the matrix A

(iv) $A(adj\ A) = (adj\ A)A = |A|I$

(v) $AdjI = I$, I is the unit matrix.

(vi) adj (AB) = (adj B)(adj A)

(vii) $|AB| = |A||B|$

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